Chapter 4 — Stoichiometry

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. One mole contains: (a) 6.02×10^{23} particles (b) 1 g particles (c) 22.4 particles (d) 6.02×10^{22} particles

2. Moles from mass: (a) n = m/M (b) n = PV (c) n = M/m (d) n = mV

3. Molar mass of water: (a) 2 g/mol (b) 20 g/mol (c) 18 g/mol (d) 16 g/mol

4. Molar volume at STP: (a) 100 dm^3 (b) 22.4 dm^3 (c) 1 dm^3 (d) 24 dm^3

5. Volume of 0.5 mol gas at STP: (a) 5.6 dm^3 (b) 44.8 dm^3 (c) 11.2 dm^3 (d) 22.4 dm^3

6. Molarity unit: (a) mol g^{-1} (b) dm^3 mol^{-1} (c) g dm^{-3} (d) mol dm^{-3}

7. Yield % formula: (a) actual - theory (b) theory/actual × 100 (c) actual/theory × 100 (d) actual x theory

8. Balanced equation gives: (a) Density (b) Atomic mass (c) Mole ratio (d) Colour

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why is the mole concept necessary in chemistry?

2. How does molar mass connect atomic mass with laboratory mass?

3. Why do equal moles of gases contain equal numbers of molecules?

4. How can gas density be used to find molar mass?

5. Why is actual yield usually less than theoretical yield?

6. How is molar concentration used in reacting solutions?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Explain the mole concept and relationships between mass, moles and Avogadro's number. **(4)**

(b) Explain molar volume, gas density and molar concentration. **(4)**

Q.4 (a) Describe the approach to stoichiometric calculations. **(4)**

(b) Explain theoretical, actual and percentage yield. **(4)**

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